Short remainder-only certificates for the greatest common divisor

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Abstract

We give a remainder-only straight-line derivation of the greatest common divisor of two positive integers using $O(\log a/\log\log a)$ remainder calls, uniformly for $a > b \geq 1$. Previously generated values may be retained and reused, but numerical constants and other arithmetic operations are not available as instructions. The construction combines a stored Euclidean prefix with three-call contractions and two smaller recursive derivations. It yields the same upper bound for the intrinsic remainder-call complexity of the gcd and of coprimeness in the Euclidean structure. This refutes the uniform-process form of the conjecture of van den Dries and Moschovakis that the Euclidean algorithm is optimal among its peers: Conjecture 3.4 of Moschovakis's 2012 notes, and the precise version (4F-5) of the Main Conjecture in his book. It does not give a deterministic remainder-only algorithm with this cost, and the deterministic conjecture remains open. The upper bound, the reduction to derivations and both refutations are formalized in Lean 4; the final section states exactly what is formalized and what is cited.

1 Statement and relation to the book

Throughout, logarithms are to base 2. A derivation starts with its two inputs. An instruction adds $\mathrm{rem}(x, y)$, where x, y are already available and $y > 0$. Earlier values remain available. No constants are supplied. Let $G(a, b)$ be the minimum number of instructions needed to make $\gcd(a, b)$ available, and let $D(a, b) = G(a, b)$ for coprime inputs. Thus $G(a, 0) = 0$ and $D(a, 1) = 0$. These are costs of generating values.

Theorem 1. *Uniformly for integers $a > b \geq 1$,*

$$G(a, b) = O\left(\frac{\log a}{\log\log a}\right).$$

In particular this bound holds for $D(a, b)$ when $\gcd(a, b) = 1$.

As usual, the asymptotic assertion concerns sufficiently large a; a bound valid at every positive input is $O(1 + \log(2 + a)/\log\log(4 + a))$.

Background. Van den Dries and Moschovakis asked whether the Euclidean algorithm is optimal among its peers, the algorithms that compute the gcd from the remainder function [3]. Moschovakis's Main Conjecture asserts that for every algorithm α from $\mathrm{rem}, \mathrm{eq}_0$ computing $\gcd(x, y)$ there is an $r > 0$ such that α makes more than $r \log x$ remainder calls on infinitely many pairs [5, p. 2].

The known lower bound is a logarithm smaller. A recursive program deciding coprimeness from $=$, $<$, $+$, truncated subtraction, iq and rem makes more than $\frac{1}{10}\log\log a$ calls on infinitely many coprime pairs [3, Theorem A]; see also [4]. The book proves it as an intrinsic bound in

$\mathbf{N}_0[\div] = (\mathbb{N}, 0, 1, =, <, +, \dot{-}, \mathrm{iq}, \mathrm{rem})$: $\mathrm{depth}_{\perp\!\!\!\perp}(\mathbf{N}_0[\div], a, b) > \frac{1}{10}\log\log a$ for infinitely many pairs [5, Theorem 6C.5, p. 187], so it holds for all uniform processes, nondeterministic ones included. A certificate in $\mathcal{N}_\varepsilon$ becomes one in $\mathbf{N}_0[\div]$ when each eq_0 or eq_1 fact is replaced by an equality with the constant 0 or 1, and its depth then exceeds its number of remainder facts by at most a constant. Hence $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) = \Omega(\log\log a)$ on infinitely many pairs. Pratt's nondeterministic algorithm [5, Theorem 2E.2 and Problems x2E.17–x2E.19] matches this bound up to a constant factor on successive Fibonacci numbers [5, Corollary 6C.7]. Whether a logarithmic bound holds for all uniform processes was left open [5, p. 188], [2, Conjecture 3.4].

In a different model, Bshouty showed that a random access machine with linear operations, shifts and indirect addressing computes the gcd in $O(\log N/\log\log N)$ operations, by reading precomputed blocks of the binary gcd algorithm from a table. He also proved that this is optimal: even with rational constants, multiplication and division by constants, and the floor function added, $\Omega(\log N/\log\log N)$ operations are necessary [1, Theorems i and ii]. The construction here uses no operation other than remainder and no table. Its saving of a factor $\log\log a$ comes instead from reusing one stored Euclidean prefix for every modulus.

The conjectures refuted. The definitions of generated certificates and intrinsic complexities are in [5, §§1D, 4C–4E] and are recalled in Section 5. Write $\mathcal{N}_0 = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0)$ and $\mathcal{N}_\varepsilon = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0, \mathrm{eq}_1)$, and $\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, f, \vec{x})$ for the least number of remainder facts in a certificate for f at $\vec{x}$ in the structure $\mathcal{A}$. Coprimeness is the relation $x \perp\!\!\!\perp y \iff x, y \geq 1 \wedge \gcd(x, y) = 1$ [5, §1A], [2]. Moschovakis's 2012 notes state:

> **Conjecture 3.4** [2]. There is a rational number $r > 0$ such that for infinitely many $a \geq b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log a$.

The book formulates one precise version of the Main Conjecture [5, p. 151]: there are a set $\mathcal{D} \subseteq \mathbb{N}^2$ and a $K > 0$ such that

$$(x, y) \in \mathcal{D} \implies \mathrm{calls}_{\mathrm{rem}}(\varepsilon, x, y) \leq L(x, y) \leq K\,\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y), \tag{4F-5}$$

where $L(x, y) = 2\log\max(x, y)$, ε is the Euclidean algorithm, and $\mathcal{D}$ satisfies the conditions (4F-3) and (4F-4). By (4F-4) the intrinsic complexity is unbounded on $\mathcal{D}$, so $\mathcal{D}$ is infinite. Dropping the first inequality and asking only for an infinite $\mathcal{D}$ gives a weaker statement, and we refute that weaker statement.

Corollary 2. *The intrinsic complexities*

$$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$$

are $O(\log a/\log\log a)$, uniformly for $a \geq b \geq 1$. Consequently Conjecture 3.4 of [2] is false, and so is the version (4F-5) of the Main Conjecture with $\mathcal{A} = \mathcal{N}_0$ or $\mathcal{A} = \mathcal{N}_\varepsilon$: for every $r > 0$, respectively $K > 0$, only finitely many pairs satisfy the stated inequality.

The corollary follows from Theorem 1 and Section 5. For (4F-5) it covers all pairs in $\mathbb{N}^2$: the gcd is symmetric, and a pair with a zero entry is certified by the single fact $\mathrm{eq}_0(0) = \mathrm{tt}$, with no remainder facts. It is formalized as `ShortGCD.not_conjecture_3_4`, `ShortGCD.not_main_conjecture_4F5` (for $\mathcal{N}_0$) and `ShortGCD.not_main_conjecture_4F5_Neps` (for $\mathcal{N}_\varepsilon$). Theorem 4F.1 [5, pp. 149–150] transfers these pointwise intrinsic bounds to a single uniform process.

The corollary settles neither the Main Conjecture for deterministic algorithms nor its form for deterministic uniform processes [2, Conjecture 3.9], [5, §4H]; see Section 6. With the lower bound

above, the intrinsic complexity of coprimeness in $\mathcal{N}_\varepsilon$ now lies between $\Omega(\log\log a)$, on infinitely many pairs, and $O(\log a/\log\log a)$, on all pairs. Closing this gap is an open problem. In particular, it is not known whether the bound of Theorem 1 is optimal for remainder-only derivations, as the corresponding bound is in Bshouty's model.

2 Normalization

Write $g = \gcd(a, b) > 0$. For positive y,

$$\operatorname{rem}(gx, gy) = g \operatorname{rem}(x, y).$$

Every generated value is divisible by g. Consequently scaling a derivation, or dividing all its values by g, preserves its length and the validity of every instruction. Hence

$$G(a, b) = D(a/g, b/g). \tag{1}$$

Define

$$T(n) = \max_{0\le b\le a\le 2^n,\ a>0} G(a, b).$$

Euclid's algorithm makes this a finite maximum. Normalization reduces an upper bound for T to coprime inputs, with equality, zero inputs, and inputs equal to one handled without calls.

3 Three-call contractions

Lemma 3. *If $m > 0$ is available and a finite collection S of available values occupies $t \ge 2$ residue classes modulo m, at most three remainder calls generate w with*

$$0 < w < \frac{m}{t-1}.$$

Proof. Sort the distinct residues. Some adjacent pair $u < v$ has gap $\delta = v - u < m/(t-1)$, because the sum of the $t-1$ adjacent gaps is less than m. Each selected residue can be generated with one call from a witness in S. If $u = 0$, use v. If $0 < u \le \delta$, use u. Otherwise $0 < \delta < u$ and $v = u + \delta$, so $\operatorname{rem}(v, u) = \delta$. This uses at most three calls. Subtraction and sorting describe the proof and the selection of instructions; they are not instructions of the derivation. □

Lemma 4 (Euclidean-prefix lemma)**.** *Let $r_0 = a > r_1 = b > \cdots > r_h > 0$ be a Euclidean prefix of a coprime pair, with $r_{i-1} = q_i r_i + r_{i+1}$ and $q_i \ge 1$. For every integer $m > 2a/r_h$, the h pairs*

$$(r_i \bmod m, r_{i+1} \bmod m), \qquad 0 \le i < h,$$

are distinct. The prefix therefore occupies t residue classes with $t^2 \ge h$.

Proof. Suppose the pairs at $i < j$ coincide. Set

$$M = \prod_{\ell=i+1}^{j} \begin{pmatrix} q_\ell & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & E \end{pmatrix}, \qquad \begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = M \begin{pmatrix} r_j \\ r_{j+1} \end{pmatrix}.$$

Right multiplication by $\left(\begin{smallmatrix} q & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends (A, B, C, E) to $(qA+B,\ A,\ qC+E,\ C)$, starting from $(q, 1, 1, 0)$ for a single factor. By induction, all entries are nonnegative, $A \ge B \ge E$, $A \ge C \ge 1$, $B \ge 1$, and $\det M = (-1)^{j-i}$. Also $Ar_j \le r_i \le a$, whence $A \le a/r_h$. Write

$$\Delta = \det(M - I) = 1 + \det M - A - E.$$

For an odd number of factors, $\Delta = -A - E < 0$. By the same recursion, a product of at least two factors has $A \geq 2$ and $E \geq 1$; thus for an even positive number of factors, $\Delta = 2 - A - E < 0$. In either case,

$$0 < |\Delta| \leq A + E \leq 2A \leq 2a/r_h < m.$$

The presumed congruence gives $(M - I)(r_j, r_{j+1})^{\mathsf{T}} \equiv 0 \pmod{m}$. Multiplication by the adjugate shows that m divides both Δr_j and Δr_{j+1}. Consecutive remainders are coprime, so Bézout's identity implies $m \mid \Delta$, a contradiction. Finally, t individual residue classes give at most t^2 ordered pairs. □

Thus a stored prefix can be reused for every available modulus $m > 2a/r_h$. When $\sqrt{h} > 1$, three calls suffice to generate

$$0 < w < \frac{m}{\sqrt{h} - 1}. \tag{2}$$

The universal quantifier over m is essential: the modulus generated by a contraction is not predetermined.

4 Construction and recurrence

Fix an integer $n > 16$ and put $N = 2^n$ and $h = \lceil \sqrt{n} \rceil$, so that $h > 4$. After normalization we may assume $a > b > 1$ and $a \leq N$. Generate the Euclidean prefix until its last positive value is at most $N^{3/4}$ or until r_h is available. Check the input b as part of this rule. The preliminary work costs at most h calls. Euclid cannot encounter zero first without having encountered the positive remainder 1.

If a positive remainder $c \leq N^{3/4}$ is encountered, generate the next remainder $s < c$ with one more call. The pair (c, s) is coprime. If $s = 0$, then $c = 1$ is already available; otherwise both inputs to the remaining derivation are at most $2^{\lceil 3n/4 \rceil}$. This gives the bound

$$h + 1 + T(\lceil 3n/4 \rceil). \tag{3}$$

Otherwise $r_h > N^{3/4}$, so $2a/r_h < 2N^{1/4}$. For $n > 12$, $N^{1/3} > 2N^{1/4}$, and Lemma 4 applies to every available $m > N^{1/3}$. Starting at $m = r_h$, repeatedly use the contraction inequality until $0 < m \leq N^{1/3}$. Set

$$\kappa_n = \sqrt{h} - 1 > 1, \qquad J_n = \left\lceil \frac{2n/3}{\log \kappa_n} \right\rceil.$$

There are at most J_n contractions: if J_n have occurred then $m < N/\kappa_n^{J_n} \leq N^{1/3}$. Since $\log \kappa_n = \frac{1}{4} \log n + O(1)$, their cost is $3J_n = O(n/\log n)$.

The small m need not be coprime to either input. Generate $u = \mathrm{rem}(a, m)$ and $v = \mathrm{rem}(b, m)$, then concatenate derivations of $d = \gcd(m, u)$ and $e = \gcd(d, v)$. These derivations exist by the definition of T, since G is symmetric in its arguments, and use only inputs bounded by m. They are not gcd-oracle instructions. The gcd identities give

$$e = \gcd(\gcd(m, a \bmod m), b \bmod m) = \gcd(m, \gcd(a, b)) = 1.$$

The cost in this case is at most

$$h + 3J_n + 2 + 2T(\lceil n/3 \rceil). \tag{4}$$

Retaining values makes concatenation legitimate. In particular, the second recursive derivation has both d and v available even if the first one uses neither original input.

The two case estimates and normalization imply

$$T(n) \le \max\{T(\lceil 3n/4 \rceil) + O(\sqrt{n}),\ 2T(\lceil n/3 \rceil) + O(n/\log n)\}. \tag{5}$$

All implied constants are independent of the inputs.

Proof of Theorem 1. For $n \ge 2$ let $f(n) = n/\log n$. The ratios

$$\frac{f(\lceil 3n/4 \rceil)}{f(n)} \longrightarrow \frac{3}{4}, \qquad \frac{2f(\lceil n/3 \rceil)}{f(n)} \longrightarrow \frac{2}{3}$$

show that, beyond a fixed threshold, $f(\lceil 3n/4 \rceil) \le \frac{7}{8} f(n)$ and $2f(\lceil n/3 \rceil) \le \frac{5}{6} f(n)$. Both overheads in (5) are at most $c_0 f(n)$ for a fixed c_0. Choose the induction threshold above 16, where the recurrence holds, and large enough that both recursive indices are at least 2 and strictly less than n. Choose $C \ge 8c_0$, increased to cover the finitely many values below the threshold. Strong induction then gives $T(n) \le Cf(n)$: the two bounds are $(7C/8 + c_0)f(n)$ and $(5C/6 + c_0)f(n)$, both at most $Cf(n)$. Finally take $n = \lceil \log a \rceil$. This also gives $G(a, b) = o(\log a)$ uniformly as $a \to \infty$. □

5 Certificates and uniform processes

Use the book's partial remainder operation: its second argument must be nonzero. Write $\mathcal{N}_0 = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0)$ and $\mathcal{N}_\varepsilon = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0, \mathrm{eq}_1)$; compare [5, (1C-10), pp. 24, 142–143]. A derivation determines a finite generated partial substructure by recording its remainder equations. Repeated equations can be discarded.

Consider first the relation $\gcd(a, b) = 1$ on all natural inputs. A derivation of 1 together with $\mathrm{eq}_1(1) = \mathrm{tt}$ certifies the relation: every common divisor of the images of the inputs divides every transported generated value, hence divides 1. Conversely, without a positive eq_1 fact the map $x \mapsto 2x$ preserves every recorded remainder and zero-test fact, as well as every negative one-test fact. It cannot preserve coprimeness. A certificate therefore reaches 1. Enumerating a finite generated diagram in generation order gives a derivation using no more calls than the number of remainder facts. Consequently, on coprime inputs,

$$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd{=}1, a, b) = D(a, b), \qquad \mathrm{calls}(\mathcal{N}_\varepsilon, \gcd{=}1, a, b) = D(a, b) + 1.$$

For the book's relation $\perp\!\!\!\perp$, which also requires $a, b \ge 1$, it suffices to add the facts $\mathrm{eq}_0(a) = \mathrm{ff}$ and $\mathrm{eq}_0(b) = \mathrm{ff}$. Conversely every certificate for $\perp\!\!\!\perp$ certifies $\gcd(a, b) = 1$. So for $a, b \ge 1$ the remainder count is unchanged, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) = D(a, b)$, and the total count is at most $D(a, b) + 3$.

For gcd as a function, the certificate must reach its output g, giving a lower bound of $G(a, b)$. For an upper bound, append

$$\mathrm{rem}(a, g) = z, \qquad \mathrm{rem}(b, g) = z, \qquad \mathrm{eq}_0(z) = \mathrm{tt},$$

where the generated z is zero. Every homomorphism π has $\pi(z) = 0$ and $\pi(g) > 0$ (by definedness of either appended remainder). Thus $\pi(g)$ divides both input images. Conversely every common divisor of those images divides $\pi(g)$ by induction over the derivation. Therefore

$$G(a, b) \le \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b) \le G(a, b) + 2.$$

Positivity facts may be appended if the function is restricted to positive inputs. For a noncoprime input, add $\mathrm{eq}_1(g) = \mathrm{ff}$ to this gcd certificate to certify noncoprimeness. Thus the sublogarithmic remainder-call bound holds for both truth values of coprimeness as well as for gcd. Theorem 4F.1 supplies a single uniform process realizing each intrinsic norm pointwise [5, Theorem 4F.1].

6 Scope and verification

Selecting a close pair of residues is uncharged existential choice in a straight-line derivation. A deterministic remainder-only algorithm cannot assume that choice is available at constant cost. Computing all prefix residues would take about h calls per contraction. The construction therefore does not settle the deterministic conjecture.

The supplied offline Python planner uses unrestricted arithmetic to select instructions; its independent replay checks operand availability, nonzero divisors, every remainder equation, and the final gcd. Below a base size (256 bits by default) it uses Euclid's algorithm. On one pseudorandom coprime pair of each size, generated from its default seed, it gives:

Input bits	Derivation calls	Euclid calls to reach 1, same pair
512	196	326
2048	669	1204
8192	2059	4754

Random pairs are not Euclid's worst case: on successive Fibonacci numbers Euclid needs about $1.44 \log a$ calls. These are checks of finite instances, not an asymptotic proof. A reproducible suite of 310 cases also passes, including 175 noncoprime pairs, with base size 64 bits and 3,543 contractions in total, and four deliberately invalid traces are rejected. The Euclidean-prefix lemma was also checked exhaustively for every coprime $a < 1500$, every prefix length, and 60 moduli above the threshold $2a/r_h$ (254,722,179 instances, no failures).

6.1 Machine-checked statements

The Lean 4.35.0 development, with mathlib pinned to version 4.35.0, proves the uniform real-logarithmic bound, its worst-case Big-O formulation, the Euclidean-prefix lemma, the complete three-call contraction lemma, normalization, and the certificate upper bounds. It contains no proof placeholders or added axioms. An axiom audit of the main results reports only the standard Lean foundations `propext`, `Classical.choice`, and `Quot.sound`. The theorem `ShortGCD.uniform_real_bound` states

$$\exists C > 0 \; \forall a, b \in \mathbb{N}, \quad 4 \le a \;\wedge\; b \le a \implies G(a,b) \le C \frac{\log_2 a}{\log_2(\log_2 a)}.$$

The proof starts from a finite-store instruction semantics in which the only instruction is remainder with a positive, available divisor. Costs are indices of derivations, rather than an assumed recurrence. A formal conversion to finite generated diagrams proves the gcd certificate bound with two extra remainder facts and the bound for positive and negative coprimeness instances. The reduction itself is also checked: on coprime inputs, the least number of remainder facts in a certificate for $\gcd(a,b) = 1$ equals $D(a,b)$ (`ShortGCD.callsRem_eq_D`), and G agrees with D (`ShortGCD.G_eq_D`).

The intrinsic complexities themselves are also defined in Lean, directly from certificates: `ShortGCD.callsCoprimeness` for the book's relation $\perp\!\!\!\perp$ in $\mathcal{N}_\varepsilon$, with positive inputs (`ShortGCD.BookCoprime`), and `ShortGCD.callsGcd` and `ShortGCD.callsRemGcd` for the gcd in $\mathcal{N}_0$ and $\mathcal{N}_\varepsilon$. Adding the facts $\mathrm{eq}_0(a) = \mathrm{ff}$ and $\mathrm{eq}_0(b) = \mathrm{ff}$ is checked to turn a certificate for $\gcd(a,b) = 1$ into one for $\perp\!\!\!\perp$ (`ShortGCD.certifiesCoprimeness_of_certifiesCoprime`), so on positive coprime pairs the complexity agrees with the reduction (`ShortGCD.callsCoprimeness_eq_callsRem`). The theorems `ShortGCD.not_conjecture_3_4`, `ShortGCD.not_main_conjecture_4F5` and `ShortGCD.not_main_conjecture_4F5_Neps` state that no $r > 0$ satisfies $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \ge r \log_2 a$ for

infinitely many $a \geq b \geq 1$, and that no $K > 0$ satisfies $2\log_2 \max(x,y) \leq K\,\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y)$ for infinitely many pairs in $\mathbb{N}^2$, for either structure. The general theory of uniform processes, Theorem 4F.1 and the lower bounds of Section 1 are cited, not formalized. Theorem 4F.1 is needed only to restate these intrinsic bounds in terms of a single best uniform process, not for the refutations themselves.

6.2 Integer parameters used in Lean

To avoid real rounding in the construction itself, the formal proof uses, for $j = \lfloor \log_2 n \rfloor \geq 32$,

$$p = \lfloor j/4 \rfloor, \quad k = 2^p, \quad h = 2^{2p} + 1, \quad J = \lfloor n/p \rfloor + 1, \quad \ell_1 = \lceil 3n/4 \rceil, \quad \ell_2 = \lceil n/3 \rceil.$$

Write $\widehat{T}(X) = \max_{a,b \leq X} G(a,b)$. The finite construction proves, for arbitrary natural parameters satisfying $k \geq 1$, $k^2 < h$, $Y \geq 1$, $2X \leq ZY$, and $X \leq Zk^J$,

$$\widehat{T}(X) \leq \max\{h + 1 + \widehat{T}(Y),\ h + 3J + 2 + 2\widehat{T}(Z)\}.$$

Taking $X = 2^n$, $Y = 2^{\ell_1}$, $Z = 2^{\ell_2}$ gives the required hypotheses. Lean checks

$$(h+1)j \leq 64n, \quad (h + 3J + 2)j \leq 64n, \quad j - 2 \leq \log_2^{\mathrm{nat}} \ell_1, \log_2^{\mathrm{nat}} \ell_2, \quad 8\ell_1 \leq 7n, \quad 16\ell_2 \leq 7n.$$

The last estimates and $8j \leq 9(j-2)$ give a factor $63/64$ for either recursive contribution after clearing denominators. Taking $C \geq 4096$ and large enough for the finite initial segment proves $\widehat{T}(2^n)\lfloor \log_2 n \rfloor \leq Cn$ for all n. Comparing integer floors and ceilings with real base-two logarithms yields the stated real bound. These parameter choices change only absolute constants, not the proof mechanism or its asymptotic order.

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